Quantum Circuit Optimisation for A5-GMR-1

Parallel conditional clocking, batched mixing, and a proof that the per-shift gate count is minimal

Technical report · revised 13 September 2026 · every figure produced by executed code in `qgmr1_code/`

1. Summary

Two new circuits are delivered. **Algorithm 5** restructures the conditional register shift so it runs wide instead of deep, cutting T-depth 4.4× at essentially unchanged gate count. **Algorithm 6** exploits a commutativity property of the 250 mixing rounds to replace 250 conditional shifts per register with 8 controlled matrix powers, cutting the T count 2.5×. Both reproduce the exact keystream of the existing Algorithm 3 and of the independent classical reference implementation. Separately, a short proof shows that no circuit can perform a single controlled shift of an L-bit register with fewer than $L - 1$ Toffolis.

×4.43	×3.22	×2.52	×4.66
T-DEPTH ALG. 5 VS ALG. 3	DW-COST ALG. 5 VS ALG. 3	T GATES ALG. 6 VS ALG. 3	T GATES, MEASUREMENT-ASSISTED (ALG. 6)

VERIFICATION STATUS — READ THIS FIRST

A 12-test suite passes, and every claim was then attacked by a **self-conducted** adversarial review built on independent re-implementations: its own cost counter, its own simulators, exhaustive synthesis checks, and a truth-table check of the lower bound (Section 7). That review found and corrected **three errors in an earlier draft of this report**, closed one test-coverage gap, and surfaced one small unfixed inefficiency (Section 8). A review by a *separate* reviewer could not be run in this session, so nothing here has yet been checked by anyone other than the author.

2. Where the cost actually is

Algorithm 3 had already removed the unconditional shifts by offset relabelling, cutting CNOTs from 76,718 to 46,382. But its T count was *identical* to Algorithm 1+2 — 181,692 in both. The conditional shift was untouched, and it dominates:

Table 1. Toffoli budget of Algorithm 3, by source.

Source	Toffolis	Share
Fredkin cascade in `cf` (the conditional shift)	**19,764**	**76%**
Feedback taps in `cf`	2,916	11%
Majorities in `c`, `unc`, `yt`	3,276	13%
Total	**25,956**	**100%**

The cascade is also *sequential*, for two independent reasons: consecutive `cswap(ctl, r[i], r[i-1])` and `cswap(ctl, r[i-1], r[i-2])` share `r[i-1]`, and every Fredkin shares the single control `ctl`. One shift therefore costs $L - 1$ Fredkin layers.

3. Algorithm 5 — parallel conditional shift

3.1 The three-reversal identity

A right-rotation by one equals two reversals:

```
rot(1)  =  rev(0 .. L−1)  then  rev(1 .. L−1)
```

Each reversal is a set of *disjoint* transpositions, so it is one layer. The two reversals use $\lfloor L/2 \rfloor + \lfloor (L-1)/2 \rfloor = L-1$ Fredkins — **exactly the same count as the cascade**. Only the shape changes. The control is fanned out over a CNOT tree into separate copies, removing the shared-control serialisation too.

BEFORE — chain, one shared control

```
6 gates, 6 time-steps

step 1 | swap(6,5) on c
step 2 | swap(5,4) on c
step 3 | swap(4,3) on c
step 4 | swap(3,2) on c
step 5 | swap(2,1) on c
step 6 | swap(1,0) on c
```

AFTER — two reversals, fanned-out controls

```
6 gates, 2 time-steps

step 1 | swap(0,6) on c0
       | swap(1,5) on c1
       | swap(2,4) on c2
step 2 | swap(1,6) on c0
       | swap(2,5) on c1
       | swap(3,4) on c2
```

Both produce `abcdefg → gabcdef`. Generated for L = 7; the real registers are L = 19, 22, 23.

Table 2. Depth of one conditional update, measured on the built circuits. Upper rows: the shift alone. Lower rows: everything `cf()` really does — three feedback-tap Toffolis, control fan-out, shift, un-fan-out.

Register	What is counted	Toffolis	T-depth old	T-depth new	Depth old	Depth new
R1 (L = 19)	shift only	18	72	8	180	20
R2 (L = 22)	shift only	21	84	8	210	20
R3 (L = 23)	shift only	22	88	8	220	20
R1 (L = 19)	**whole conditional update**	**21**	**84**	**20**	**204**	**50**
R2 (L = 22)	**whole conditional update**	**24**	**96**	**20**	**234**	**50**
R3 (L = 23)	**whole conditional update**	**25**	**100**	**20**	**244**	**50**

The shift alone becomes 11× shallower. The **whole update becomes about 4–5× shallower**: the three feedback taps all target the same qubit and must still run in sequence (T-depth 12), and the fan-out tree adds CNOT layers. *An earlier draft quoted only the 11× figure; the honest per-update figure is 4–5×*. The whole-circuit results in Table 5 were always computed on the real circuit and are unaffected.

3.2 Majority with one Toffoli

Using $\mathrm{MAJ}(a,b,c) = a \oplus ((a \oplus b) \wedge (a \oplus c))$, a majority costs 1 Toffoli + 5 CNOT instead of 3 Toffolis. Applied to the clock-control majority and the three output majorities, this takes 3,276 Toffolis to 1,092 — a saving of exactly 2,184, which is the *entire* difference between Algorithm 3's 25,956 and Algorithm 5's 23,772.

> **BE PRECISE ABOUT THIS IN THE PAPER**
>
> The conditional clocking lost **zero** gates: 19,764 Fredkins before, 19,764 after. It became shallower (11× for the shift, 4–5× for the whole update). The 8% gate reduction of Algorithm 5 comes entirely from the majority rewrite. Conflating the two would be wrong.

4. Algorithm 6 — batched mixing

4.1 The observation

During the 250 mixing rounds `y(t)` is never called, so R1/R2/R3 are read by **nothing except their own `cf`**. No intermediate state of those registers is ever observed. Each is a self-contained conditionally-clocked LFSR, and all 250 of its steps are powers of the same matrix M_k, so they **commute**:

```
250 conditional clocks of R_k   ==   M_k ^ (number of set clock bits)
```

Both halves were checked: a gate-level audit of 18,088 mixing-round gates over 68 offset combinations found no gate that combines R_k data with anything but R_k, its clock bit, and its own control copies; and the identity held on 600 random (state, 250-bit clock pattern) trials.

4.2 The construction

1. Run R4 forward 250 steps (unconditional, deterministic) and record each clock bit $c_k(t)$ in its own qubit. Cost: 2 Toffolis per round.
2. Sum the 250 bits with a **carry-save tree** into an 8-bit number n_k: 244 Toffolis and 244 carry ancillas per register.
3. Apply $M_k^{n_k}$ with a barrel of controlled $M_k^{2^w}$, w = 0..7, each synthesised as a GF(2) linear circuit with every CNOT promoted to a Toffoli.

Table 3. Algorithm 6 Toffoli budget, counted phase by phase on the built circuit (independently recounted).

Phase	Component	Toffolis
Mixing	clock-bit majorities (250 × 2)	500
	carry-save trees (3 × 244)	732
	barrels of controlled matrix powers	3,281
	subtotal (was 18,000 in Algorithm 5)	**4,513**
Keystream	74 rounds × 78, unchanged from Algorithm 5	5,772
Total		**10,285**

> **BETTER THAN PROJECTED**
>
> The clock accumulator was estimated at ~3,400 Toffolis before it was built. Measured: **1,232** (500 + 732). That is why Algorithm 6 landed at 10,285 rather than the projected 12,918.

5. Lower bound: one controlled shift needs $L - 1$ Toffolis

Claim. Every circuit of NOT, CNOT and Toffoli gates that implements a controlled cyclic shift of L bits contains at least $L - 1$ Toffolis — with any number of ancillas, garbage allowed, and Toffolis nested arbitrarily. Fredkins are counted as one Toffoli each.

Proof. A Toffoli XORs one product of two current wire values into its target; NOT and CNOT are affine. By induction, at every moment every wire is an affine combination of the inputs and of the products $P_1, \ldots, P_m$ created so far. So the outputs lie in span{1, inputs, $P_1, \ldots, P_m$}, which forces $m \geq d$, where d is the dimension of the span of the outputs modulo affine functions. Output i of the controlled shift is $s_i \oplus ctl \cdot (s_i \oplus s_{i-1})$. A combination $\Sigma\, c_i f_i$ is affine exactly when $c^T(M \oplus I) = 0$, so $d = \text{rank}(M \oplus I) = L - 1$. ■

Table 4. The dimension d, computed two independent ways, and the Toffolis actually used.

Register	rank($M_{cyclic} \oplus I$)	d from full truth table	Fredkins used	d for full LFSR clock	Toffolis used
R1 (L = 19)	18	18	18	19	21
R2 (L = 22)	21	21	21	22	24
R3 (L = 23)	22	22	22	23	25

The truth-table computation (2^{24}-entry tables for R3) agrees with the matrix rank for every L from 2 to 12 and for the three real registers. As a check on the argument itself, an exhaustive search for $L = 3$ found that none of the 1,024 possible one-Toffoli circuits works, while a two-Toffoli circuit exists — the bound is tight.

What this does and does not say. $L - 1$ Fredkins is the minimum for *one* controlled shift, which is why Algorithm 5 could only win on depth. It says nothing about 250 shifts composed — a different function, which is exactly the slack Algorithm 6 exploits. The last two columns give a proven floor of L Toffolis for the full controlled LFSR clock (feedback taps included) against $L + 2$ used: a possible saving of 2 per update, for which no construction is known.

6. Results

Table 5. Anand et al. convention: gate set NOT + CNOT + Toffoli, 1 Toffoli = 7 T, full parallelism. Every cell independently recounted.

Circuit	Toffoli	T gates	Clifford	T-depth	Full depth	Width	DW-cost
Alg. 1+2	25,956	181,692	286,349	40,176	111,306	231	25,711,686
Alg. 3 (prior best)	25,956	181,692	256,013	40,184	96,110	231	22,201,410
Alg. 4 (dagger)	24,318	170,226	239,549	31,128	76,379	1,815	138,627,885
Alg. 5 (parallel)	**23,772**	**166,404**	**265,385**	**9,076**	**26,125**	**264**	**6,897,000**
Alg. 6 (batched)	**10,285**	**71,995**	**113,327**	**9,164**	**23,579**	**1,746**	**41,168,934**

Algorithm 6 beats Algorithm 4 in every column — Toffoli, T, Clifford, T-depth, full depth, width (1,746 vs 1,815) and DW-cost. Both are oracle-only. **Algorithm 5 remains the best narrow circuit**: its DW-cost of 6.9M is the lowest of all five, and it is the only new circuit that leaves every scratch qubit clean.

Table 6. Separate convention — measurement-assisted Toffolis. NOT comparable with Table 5.

Circuit	AND pairs	open AND	Jones Toffolis	T gates	T-depth †	Measurements †
Alg. 5	546	0	22,680	92,904	1,945	23,226
Alg. 6	**546**	**732**	**8,461**	**38,956**	**1,968**	**9,007**

Rules: a Gidney AND costs 4 T to compute and 0 T to erase; a Jones Toffoli costs 4 T. Both figures are **confirmed against the published abstracts** (Gidney, Quantum 2, 74, 2018; Jones, Phys. Rev. A 87, 022328, 2013). † The T-depth column assumes T-depth 1 per Jones Toffoli and per AND compute. The measurement column assumes one measurement per AND erase and per Jones Toffoli, and none per AND compute. *Those assumptions were not confirmed from the sources*, which describe the constructions only in their abstracts. The Clifford column is omitted deliberately, since those counts are not sourced. Measurement-assisted constructions sit outside Table 5's gate set, so this is a separate, footnoted row.

Relative to Algorithm 3's 181,692 T: Algorithm 5 measured gives ×1.96, Algorithm 6 measured gives **×4.66**.

7. Verification performed

Table 7a. Test suite `verification/test_algorithms_5_6.py` — 12 tests, 13 checks, ~70 s.

#	Check	Scope	Result
1	Three-reversal shift == Fredkin chain	L = 19, 22, 23, all basis states	PASS
2	Fan-out copies control, restores copies to |0⟩	1..12 copies, both control values	PASS
3	Alg. 5 keystream == Alg. 3, scratch clean	10 key/frame pairs	PASS
4	Gidney AND pair preconditions, Alg. 5	546 pairs × 4 keys	PASS
5	`step_phys` == the real `cf` circuit	all offsets × basis states	PASS
6	250 conditional clocks == M^popcount	200 trials per register	PASS
7	Alg. 6 keystream == Alg. 3	10 key/frame pairs	PASS
8	Alg. 5 and Alg. 6 == **classical reference**	10 pairs (20 in an earlier run)	PASS
9	Barrel == repeated clocking for **every** count	n = 0..255, 3 registers	PASS
10	Carry-save popcount exact	6,692 patterns, exhaustive to 10 bits	PASS
11	Inside the real Alg. 6: 30 trees output true counts; AND tags valid	10 keys, one simulation each	PASS
12	Cost categories partition every Toffoli	Alg. 5 and Alg. 6	PASS

Table 7b. Self-conducted adversarial review, each part written independently of the code it checks.

Attack	What was run	Result
Cost numbers	own flattener and ASAP layering; all 28 cells of Table 5 and the Table 3 phases	MATCH
GF(2) synthesis	all 20,335 invertible matrices up to 4×4 on every input; 760 random up to 23×23; forward-order emission confirmed to fail	PASS
Controlled barrel	1,408 qiskit cases: real M^{2^w}, control 0 and 1	PASS
Reversal identity	L = 2..30; all 2^L states up to L = 14	PASS
No reads during mixing	18,088 gates, 68 offset combinations, Alg. 3 and Alg. 5	PASS
Hidden global state	rebuilt after other builds with deliberately corrupted globals; gate lists identical	PASS
Lower bound	truth-table dimension for L = 2..12 and real registers; exhaustive L = 3 search	PROVEN
Literature figures	published abstracts: 4 T / 0 T / 4 T confirmed; T-depth and measurement details not	PARTIAL
Coverage of real keys	clock counts from 10 keys fall in [173, 200]: bit 7 never toggles, now covered by test 9	CLOSED
Review by a separate reviewer	not available in this session	NOT RUN

8. Errors found, and what is not established

Errors found and corrected

1. **Measurement over-count.** `profile_measured()` charged a measurement to each open AND compute, which uses none. Algorithm 6's count was 9,739; the correct figure is **9,007**. Fixed in code; test 12 guards it.

2. **Overstated depth gain.** An earlier draft called the conditional update 11× shallower. That is true of the shift alone; the whole update, including feedback taps and fan-out, is 4–5× (Table 2). The whole-circuit numbers were never affected.
3. **Understated lower bound.** An earlier draft called the $L - 1$ bound "a strong argument needing a citation". It is now proven (Section 5).
4. **Two test defects**, found while building (8 September): a Gidney check that misfiled erases as computes, and a test case using a 20-bit frame number that the reference implementation rejects. Both were the tests' faults, not the circuits'.
5. **Coverage gap.** Real keys never exercise the control-0 branch of controlled-M^{128}. Test 9 now checks every count from 0 to 255.

Found and not fixed

Every conditional update fans the control out to 11 copies, but R1's reversals use only 9. That wastes 4 CNOTs per R1 update: 1,296 CNOTs in Algorithm 5 and 296 in Algorithm 6, plus 2 idle qubits. It has no effect on correctness, Toffolis, T gates or T-depth. It is a three-line change, left for a deliberate follow-up because it alters published Clifford and width figures.

> **NOT ESTABLISHED**
>
> 1. **No separate reviewer.** The adversarial review above was self-conducted.
> 2. **Table 6's T-depth and measurement columns** rest on per-construction figures that were not confirmed from the sources.
> 3. **Full-cipher equivalence is sampled.** Components are checked exhaustively, but the complete circuits are compared on 20 of the 2^{83} key/frame inputs.
> 4. **Algorithm 6 is oracle-only.** It leaves ~1,482 qubits dirty for ENC† to clear, and its DW-cost is worse than Algorithm 5's. That its open ANDs remain valid when erased inside ENC† is argued, not simulated.
> 5. **The barrel is not optimally synthesised.** Plain Gauss–Jordan was used; Patel–Markov–Hayes synthesis would plausibly remove a further 20–30 % of its 3,281 Toffolis.

9. Files delivered

File	Contents
`algorithm5_a5gmr1_par.py`	three-reversal conditional shift, control fan-out, one-Toffoli majority, Gidney AND role tags
`algorithm6_a5gmr1_batch.py`	batched mixing: clock-bit record, carry-save popcount, barrel of controlled matrix powers
`gf2_synth.py`	GF(2) matrices, self-checking Gauss–Jordan CNOT synthesis, carry-save tree planner
`resources.py`	`profile_measured()` for the measurement-assisted convention (with the measurement fix)
`verification/test_algorithms_5_6.py`	the 12 tests of Table 7a; run a subset with `python test_algorithms_5_6.py 9 10 11`

10. Suggested next steps

1. **Have someone else review it** — the largest remaining gap in confidence.
2. **Confirm the Jones and Gidney T-depth and measurement figures** from the full papers before using Table 6.

3. **Fix the R1 fan-out** (11 → 9 copies) and regenerate the Clifford and width columns.
4. **Swap Gauss–Jordan for Patel–Markov–Hayes** in the barrel synthesis.
5. **Build a narrow Algorithm 6** that uncomputes the trees in chunks, aiming for a DW-cost below Algorithm 5's.

Revised 13 September 2026. All figures were produced by executing the code described. The adversarial review was conducted by the author; no separate reviewer has checked this work. Sources: quantum-journal.org/papers/q-2018-06-18-74 · link.aps.org/doi/10.1103/PhysRevA.87.022328